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A deep learning-based IDS for IoT: model proposal and comparative study of dataset balancing techniques

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ABSTRACT

Intrusion detection in Internet of Things (IoT) environments presents challenges due to the diversity of connected devices and their resource limitations. IoT networks generate complex, imbalanced traffic where benign activity predominates over attack instances. This imbalance hampers the performance of traditional intrusion detection systems, which struggle to generalize effectively. In this study, we present a deep neural network-based system that leverages advanced data balancing techniques – such as subsampling, Synthetic Minority Over-sampling Technique (SMOTE), and Tomek Links – combined with cross-validation to enhance the model's generalization and minimize overfitting. Evaluations on CICIDS2017, UNSW-NB15, and BoT-IoT datasets showed accuracy rates of 99.2%, 99.7%, and 99.8%, respectively. These results demonstrate that our methodology outperforms traditional models, especially in detecting minority attack classes, which were previously challenging due to data imbalance. The use of data balancing and cross-validation significantly improved model stability and sensitivity to diverse attack scenarios. Our findings suggest that incorporating these techniques can substantially enhance the security of IoT environments, providing a robust approach for differentiating between normal and malicious activities, thus contributing to more reliable and scalable intrusion detection systems.

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IoT; cybersecurity; intrusion detection; IDS; artificial intelligence

1. Introduction

The expansion of the IoT has marked a milestone in modern technology, allowing billions of devices to collect and share data in real time without human intervention. This growth is driven by its integration into key areas such as healthcare, smart cities, industry, and smart homes, where IoT devices facilitate efficiency, automation, and connectivity at various scales. According to estimates, the number of IoT devices is expected to reach 75 billion by 2025 (Alam, 2018), reflecting significant growth compared to approximately 8 billion connected devices in 2017.

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The increasing number of connected devices has significantly expanded the attack surface in IoT environments. The diversity and limited computational resources of these devices create a highly vulnerable environment, making security a major challenge in this field (Tariq et al., 2023). Many IoT devices lack sufficient processing power and storage capacity, restricting their ability to implement advanced security measures. This makes them attractive targets for attackers, as most lack robust authentication or encryption systems, making system vulnerabilities easier to exploit (Tawalbeh et al., 2020; Williams et al., 2022).

IoT devices are particularly vulnerable due to several inherent limitations. Communication within IoT environments relies on protocols that prioritize efficiency over security, such as Message Queuing Telemetry Transport (MQTT), Constrained Application Protocol (CoAP), and Routing Protocol for Low-Power and Lossy Networks (RPL). These protocols are susceptible to data interception and manipulation, as they often lack advanced encryption and proper access controls, making data confidentiality and integrity vulnerable to attacks.

Another major issue is the lack of a comprehensive security management infrastructure, complicating the deployment of necessary security patches or updates. Without secure or automated update capabilities, IoT devices, once deployed, are challenging to protect from emerging threats. This limitation means that compromised devices can remain vulnerable for extended periods, posing significant risks to the entire network.

In response to these challenges, IoT requires security strategies tailored to its limitations, leading to the adoption of specific defense technologies, including Intrusion Detection Systems (IDS) (Bakhsh et al., 2023). IDS have emerged as an essential response for identifying and mitigating threats in IoT networks, as they can analyse data traffic and detect suspicious behaviours or unusual patterns in real time. This capability is crucial for protecting both the integrity (ensuring data has not been altered) and confidentiality (ensuring only authorized users access the information) of IoT networks.

In IoT networks, IDS are particularly valuable because these networks are constantly expanding and have a continuous data flow. IDS are primarily divided into two types (Einy et al., 2021): signature-based IDS and anomaly-based IDS. Signature-based IDS compare network traffic against a database of known threat patterns or ‘signatures,’ which are specific sequences of code or behaviour that identify a known attack. This approach is effective in detecting previously cataloged attacks, but its major limitation lies in its inability to recognize unknown threats or new variants of known attacks, such as zero-day attacks, which exploit vulnerabilities that have yet to be discovered or patched by developers.

On the other hand, anomaly-based IDS use statistical or machine learning (ML) techniques to create models of normal behaviour and identify deviations that might indicate suspicious activity. This method has proven to be more adaptive and suitable for IoT environments, which are inherently dynamic and diverse. By establishing a baseline profile of ‘normal’ activity, anomaly-based IDS can detect anomalies essential for identifying new threats. However, they face the ongoing challenge of minimizing false positives (FP), which are alerts triggered by legitimate activities incorrectly identified as malicious, leading to excessive alerts and increased resource consumption.

Intrusion detection systems can be adapted based on their location within the network, resulting in two main approaches: Network-Based IDS (NIDS) and Host-Based

IDS (HIDS). Network-Based IDS are designed to monitor the traffic passing through the network, identifying suspicious activity at the level of data packets moving between devices. Their ability to analyse data flows across multiple nodes makes them particularly useful in environments like IoT, where devices often communicate extensively. In such a scenario, a well-designed NIDS can effectively monitor large volumes of data in real-time, detecting attack patterns without burdening the resources of individual devices.

In contrast, Host-Based IDS are implemented on individual devices or nodes, monitoring suspicious activities directly within the system, such as unauthorized access attempts or unexpected changes in critical files (Satilmis et al., 2024). While HIDS provide in-depth analysis on a per-device basis, their implementation in IoT environments can be challenging due to the limited processing power and storage capacity of most IoT devices. Given these constraints, NIDS are often considered a more suitable solution for IoT networks, as they offer efficient detection of network-level threats with minimal impact on device performance and energy consumption.

With the advancement of artificial intelligence (AI) and deep learning (DL), more sophisticated models such as Deep Neural Networks (DNN) have been introduced. DNN enables more precise and real-time detection by analyzing large volumes of data and detecting complex patterns (Bakhsh et al., 2023). Models such as Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks are increasingly common, as they combine the ability to analyse sequential data with the capacity to capture spatial features. These advances have allowed IDS to evolve into systems that not only detect known attacks but also learn and continuously adapt to new threats in the IoT context.

With this perspective, our research focuses on designing an intrusion detection system based on deep neural networks (DNN) that optimizes detection in IoT environments, adapting to the resource limitations of devices and anticipating emerging threats. The main contributions of this work include the proposal of an efficient IDS for IoT environments, leveraging advanced data balancing techniques, such as undersampling, SMOTE, and Tomek Links, to enhance the detection of minority classes. Additionally, we evaluated the effectiveness of the system using three benchmark datasets (CICIDS2017 ("[CICIDS2017 Dataset](#)"), UNSW-NB15 ("[UNSW-NB15 Dataset](#)"), and BoT-IoT ("[BoT-IoT Dataset](#)"), demonstrating its capacity to generalize across different IoT scenarios. We also implemented K-Folds cross-validation to maximize the model's generalization capability and reduce overfitting. Finally, a comparative analysis of the results is presented, showing how the proposed approach significantly improves the detection of infrequent attacks compared to traditional techniques, which is crucial for IoT network security.

The remainder of the document is organized as follows: Section II addresses the state of the art in IDS for IoT, highlighting the main techniques and recent approaches in literature. Section III provides a detailed description of the methodology employed in our proposal, including the technical aspects of the deep neural network-based model and data preparation. Section IV presents the experiments conducted, allowing for replication of the study. Section V details the results obtained, evaluating the effectiveness and accuracy of our solution. Finally, Section VI discusses the conclusions drawn and proposes potential future research directions to enhance security systems in IoT.

2. Related works

Intrusion detection in IoT is challenging due to the heterogeneity and resource limitations of the devices involved. Below is a detailed analysis of the most recent and representative approaches, detailing the techniques used, limitations, and results obtained.

DL techniques, such as LSTM, CNN, Transformers, and Generative Adversarial Networks (GAN), have been widely explored for intrusion detection in IoT due to their ability to identify complex patterns in large data volumes. LSTM-based models have shown promising results in capturing temporal dependencies in network traffic. For instance, Almiani et al. (2020) employed an LSTM-based model using the UNSW-NB15 dataset, achieving 99.7% detection accuracy for attacks such as Distributed Denial of Service (DDoS). LSTM's capacity to recognize temporal dependencies is particularly crucial for identifying anomalous behaviour that evolves over time. However, one of the main limitations of LSTM is its high computational demand, which complicates implementation in resource-constrained IoT devices.

To address both temporal and spatial complexities, some studies have opted for more sophisticated architectures that can handle these challenges more effectively. Ullah et al. (2023) combined Gated Recurrent Units (GRU) with Transformers, achieving 99.5% accuracy on the BoT-IoT dataset and significantly reducing FP compared to models based solely on LSTM or GRU. The Transformers' attention mechanism helps analyse intricate relationships in network traffic by focusing on specific sequences to identify suspicious activities. Nevertheless, the high computational requirements still pose significant challenges for practical implementation in IoT contexts.

Another effective approach to enhance efficiency, particularly in IoT environments, is the use of CNN. CNNs are advantageous due to their automatic extraction of key features with reduced computational burden, making them suitable for real-time applications. Eskandari et al. (2020) leveraged CNN for intrusion detection in edge devices, achieving early threat detection with high accuracy and minimal FP. Their system was tested in a home automation environment, detecting attacks such as Port Scanning and Hypertext Transfer Protocol (HTTP) brute force. Similarly, Soliman et al. (2023) developed a CNN-based model optimized for industrial IoT devices using the ToN_IoT dataset ("Ton_IoT Dataset"), achieving 99.99% accuracy in binary classification and 99.98% in multiclass classification. These approaches demonstrate that well-optimized CNNs can be powerful yet resource-efficient solutions for real-world IoT applications.

The detection of rare and sophisticated attacks, such as zero-day attacks, requires combining different deep learning architectures. Li et al. (2024) proposed combining CNN-LSTM with GAN to detect DoS attacks in IoT networks. GANs generated synthetic malicious data and built baselines for legitimate behaviour, allowing the model to detect previously unseen attack vectors. This semi-supervised approach achieved over 98% accuracy on datasets like NSL-KDD ("NSL-KDD"), CSE-CIC-IDS2018 ("CSE-CIC-IDS2018"), and BoT-IoT while also reducing FP significantly. However, the computational complexity of this combined architecture limits its suitability for low-resource environments, highlighting a common issue for such hybrid approaches.

Attention mechanisms have also been applied to enhance the ability to detect intricate and subtle attack behaviours. Swathi and Bindu (2024) developed a bidirectional autoencoder model with attention to pinpoint complex patterns related to intrusion. This system

achieved 99.71% accuracy on the UNSW-NB15 dataset and also performed well on NSL-KDD with 98.97%. The addition of an attention mechanism allowed the model to focus on critical features within the data, which significantly improved its detection capabilities. However, the higher computational requirements remain an obstacle to broader applicability in lightweight IoT devices.

Similarly, Malhotra et al. (2016) employed an LSTM Autoencoder approach using the CICIDS2017 dataset, which proved useful for reconstructing sequences and identifying abnormal deviations indicating intrusion. However, similar to other LSTM models, the need for significant memory and processing power remains a limiting factor in practical IoT deployments.

Optimization techniques, such as Genetic Algorithms (GA) and Whale Optimization Algorithm (WOA), have also been explored to improve the efficiency of intrusion detection systems for IoT devices. Osman et al. (2024) used GA for feature selection in the RPL-ELIDS dataset, reducing features from 113 to 17, which enhanced the model's efficiency with a 97.9% accuracy. By selecting the most informative features, GA reduces dimensionality and computational overhead. However, GA, when combined with DNN, may increase training time due to the iterative nature of the feature optimization process. Similarly, Ramana et al. (2022) combined GRU with WOA to optimize hyperparameters, achieving a remarkable 99.85% accuracy. These optimizations improve both accuracy and detection speed, yet their practicality in low-resource devices still faces challenges due to computational complexity.

In the context of anomaly detection, advanced models like Nie et al. (2024) multi-view Transformer approach have demonstrated enhanced capabilities in analyzing network traffic from different temporal and semantic perspectives. This multi-view technique, which achieved high accuracy in detecting anomalous behaviour, underscores the value of combining spatial, temporal, and content-based insights. However, the computational cost makes it difficult to apply such an approach to the limited-resource environments typical of IoT devices.

Unsupervised or semi-supervised learning techniques, such as Variational Autoencoders (VAE), have proven to be powerful tools in situations with unlabeled data. Guo et al. (2020) developed a model based on Gaussian Mixture Model (GMM), VAE, and GRU to detect anomalies in multidimensional time series data from the SWaT dataset ("SWaT Dataset"). This combination led to a 47.88% improvement in F1-score, allowing better identification of anomalous activities in critical infrastructure. Such methods are well-suited for characterizing the diverse and multimodal nature of IoT network traffic, though they are still hindered by substantial computational requirements.

Federated Learning (FL) offers an innovative path to improving the efficiency and privacy of IDS in IoT. Attota et al. (2021) applied FL combined with Grey Wolf optimization on the BoT-IoT dataset, achieving better detection accuracy while enhancing data privacy by training models locally and avoiding the need to transfer sensitive information. This approach is especially promising for applications requiring high levels of data privacy, such as healthcare and personal monitoring devices.

In general, research in IDS for IoT has focused on optimizing models for greater efficiency, particularly through the development of lightweight and adaptive models. While DL techniques have shown promising accuracy, they often face implementation challenges due to high resource demands. On the other hand, optimization approaches

such as GA, WOA, and FL present attractive solutions for reducing complexity, improving detection, and maintaining privacy, thus enhancing the overall viability of IDS in IoT environments.

However, the conclusion of the state of the art reveals some significant gaps. Despite recent advances, most existing solutions face substantial limitations when implemented on devices with extremely limited resources, which is common in many IoT environments. Moreover, many models fail to adequately balance accuracy in detecting minority classes with resource efficiency, a critical aspect for the security of IoT networks. Our research addresses these gaps by proposing a DNN-based IDS optimized for IoT environments. This system not only integrates advanced data balancing techniques such as SMOTE and Tomek Links to enhance minority class detection but also adopts cross-validation to improve the model's generalization capability.

In addition, our study delves into the impact of data preprocessing on model performance, evaluating three different configurations: using the original dataset, undersampling the majority class, and combining undersampling with SMOTE and Tomek Links. We have analysed how these configurations affect the model's ability to effectively detect intrusions. During model training, we also compared the use of cross-validation with the absence of this technique, highlighting how both strategies influence the model's generalization and prevent overfitting. Thus, our work not only presents an effective DNN but also provides detailed insights into how different preprocessing and validation strategies impact threat detection in IoT networks.

3. Proposed approach

In this section, the approach developed for intrusion detection in IoT environments is explained, divided into two subsections. The first focuses on the development of a DNN-based model, describing the model architecture and optimization techniques employed to enhance intrusion detection in IoT networks. The second addresses the selection of datasets used, analyzing their relevance within the context of cybersecurity networks.

3.1. DNN model

This subsection presents the architecture of the DNN model developed for intrusion detection in IoT environments. The choice of this type of neural network is characterized by multiple hidden layers that enable the modelling of complex relationships between data features. In the IoT context, where traffic is highly varied and may include both normal and malicious patterns, a DNN-based model is suitable because it can capture these complex relationships and learn subtle patterns from the available data.

DNNs are particularly useful in this domain due to their ability to generalize well when handling large volumes of data. Thanks to their multiple layers, these networks can extract low-level features – basic attributes of network traffic such as connection duration, the number of packets sent or received, and response times – in the initial layers. These features are then combined at higher levels, allowing the model to learn more abstract and complex representations of traffic behaviour, which is crucial for detecting anomalous patterns and potential attacks.

Compared to other techniques such as LSTM, CNN, or Transformers, DNNs offer a significant advantage in terms of computational efficiency. While LSTM and Transformers are powerful for capturing temporal relationships in data sequences, their implementation in IoT devices may be impractical due to their high processing and memory requirements. In contrast, DNNs have lower computational complexity and are easier to deploy in resource-constrained environments, making them ideal for use in this context.

It is recognized that DNNs also have limitations, particularly concerning overfitting when working with imbalanced data. Overfitting occurs when a model fits too closely or even exactly to its training data, resulting in a model that cannot make accurate predictions or inferences on any data other than the training set. To address these limitations, specific techniques were incorporated to improve the model's generalization capacity.

The developed model comprises three hidden dense layers with 256, 128, and 64 neurons respectively, following a pyramidal architecture that progressively reduces dimensionality. This design allows the network to learn hierarchical feature representations: the first layer captures low-level traffic characteristics, the second layer combines these into intermediate patterns, and the final layer produces compact high-level representations suitable for classification. This architecture balances model capacity with computational efficiency, making it suitable for deployment in resource-constrained IoT environments while maintaining sufficient representational power for complex intrusion detection tasks.

The architecture begins with a dense layer of 256 neurons using a Rectified Linear Unit (ReLU) activation function. An activation layer applies a mathematical function to a layer's output, introducing non-linearity to allow the model to learn complex relationships. Dense layers are essential for learning complex, non-linear representations of data, and ReLU activation is widely used due to its simplicity and effectiveness in capturing non-linear patterns without introducing issues such as vanishing gradients, which are common in other activation functions like sigmoid or tanh.

Adding more neurons in a layer means the model has greater capacity to learn complex data features but also increases the risk of overfitting and computational load. Each neuron in a layer applies a transformation to its inputs, learning a specific representation of the pattern in the data, and contributes to the model's decision-making by connecting with other neurons in successive layers. This first layer helps the model understand and process initial network traffic features that may indicate anomalous behaviour.

Following this dense layer, Batch Normalization was implemented to normalize the output of the previous layer. This normalization stabilizes the input to the next layer, keeping values within a stable range, which helps stabilize and accelerate the training process. This process makes the model less sensitive to weight initializations and supports more robust learning. By using Batch Normalization, extreme activation values that could negatively impact training are avoided, helping to improve model convergence.

Next, Dropout is applied with a rate of 30% to reduce overfitting. Dropout randomly deactivates some neurons during training, helping the model avoid dependence on specific combinations of features and learn more general patterns. The 30% rate was chosen as an appropriate balance to avoid overfitting without compromising the model's learning ability. This rate is commonly recommended in literature, yielding good results in similar problems. In IoT contexts with often imbalanced data, Dropout

improves the model's generalization to new data. After the first layer, the model continues with a dense layer of 128 neurons, followed by another Batch Normalization and Dropout layer. Then, a 64-neuron layer is added, with the same normalization and regularization pattern. These successive layers allow the model to learn intermediate-level features – features representing more complex patterns than the initial basic features, though not as abstract as the final output – and finish with a compact and rich representation of network traffic, crucial for differentiating between benign and malicious traffic.

Finally, the model concludes with an output layer with as many neurons as classes in our classification problem, representing the number of classes evaluated in the dataset. The output layer's activation function is Softmax, which converts outputs into probabilities, enabling us to classify network traffic into different categories.

This modular and well-regularized architecture enables the model to capture complex traffic patterns, effectively distinguishing between multiple types of threats. [Figure 1](#) provides a graphical representation of the developed architecture, highlighting the distribution of layers and the connections between neurons.

For model training, we employed the Adam optimizer with an initial learning rate of 0.0001. Adam was selected for its adaptive learning rate capabilities and computational efficiency, while the relatively low learning rate ensures stable convergence and prevents overshooting optimal solutions, which is particularly important when dealing with imbalanced datasets. The model was trained using sparse categorical cross-entropy as the loss function, which is specifically designed for multi-class classification problems and efficiently handles integer-encoded labels without requiring one-hot encoding, thereby reducing memory requirements. Training was conducted with a batch size of 1024, chosen to leverage GPU parallelization capabilities while balancing between gradient estimate stability and memory constraints. The maximum number of epochs was set to 50, although

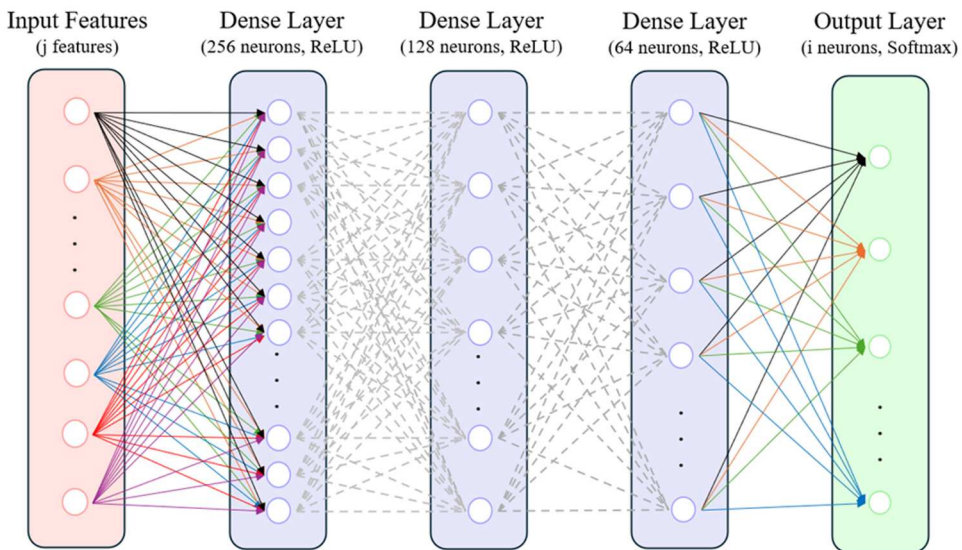


Figure 1. DNN model architecture. Intrusion classification in IoT environments is achieved using a DNN with three dense layers for feature extraction and a Softmax output layer for classification.

early stopping typically halted training before reaching this maximum, preventing overfitting by monitoring validation loss with a patience of 5 epochs.

Additionally, techniques were incorporated to ensure effective training and address challenges like imbalanced data and overfitting. First, class weights were calculated using the `compute_class_weight` function from sklearn. This technique assigns a higher weight to minority classes to ensure that the model pays sufficient attention to attack examples during training. This compensates for the difference in sample quantity for each class, enhancing the model’s ability to detect intrusions.

Various callbacks were also added to improve training efficiency and further mitigate overfitting issues. An Early Stopping callback was used, which monitors validation loss during training and halts the process if the loss does not improve after a certain number of consecutive epochs. This approach prevents over-training of the model beyond the optimal point, thereby avoiding the learning of noise or dataset-specific features that may not generalize well to new data.

ReduceLROnPlateau callback was implemented too, which automatically decreases the learning rate if the validation metric does not improve over a set number of epochs. This enables the model to fine-tune itself in the later training stages, helping it escape local optimization points and promoting better convergence.

Table 1. DNN architecture and training hyperparameters.

	Component	Configuration	Rationale
Architecture	Input Layer	Variable (dataset-dependent)	Matches feature dimensionality after preprocessing
	Hidden Layer 1	256 neurons, ReLU	Captures low-level traffic features with sufficient capacity
	Batch Normalization 1	Default parameters	Stabilizes training and accelerates convergence
	Dropout 1	Rate = 0.30	Prevents overfitting; standard rate from literature
	Hidden Layer 2	128 neurons, ReLU	Learns intermediate-level feature combinations
	Batch Normalization 2	Default parameters	Maintains activation stability between layers
	Dropout 2	Rate = 0.30	Consistent regularization across architecture
	Hidden Layer 3	64 neurons, ReLU	Produces compact high-level representations
	Batch Normalization 3	Default parameters	Ensures stable inputs to output layer
	Dropout 3	Rate = 0.30	Final regularization before classification
Training Parameters	Output Layer	N neurons (class-dependent), Softmax	N = number of classes; Softmax for probability distribution
	Optimizer	Adam	Adaptive learning rates; computationally efficient
	Initial Learning Rate	0.0001	Ensure stable convergence on imbalanced data
	Loss Function	Sparse Categorical Cross-Entropy	Efficient for multi-class with integer labels
	Batch Size	1024	Balances GPU efficiency and gradient stability
Regularization	Maximum Epochs	50	Sufficient for convergence with early stopping
	Class Weights	Balanced (sklearn)	Compensates for class imbalance
	Early Stopping	Patience = 5, monitor val_loss	Prevents overfitting; restores best weights
	ReduceLROnPlateau	Factor = 0.5, patience = 3	Fine-tunes learning in later training stages

Table 1 summarizes the complete architecture specifications and training hyperparameters, providing all details necessary for exact replication of our experimental setup.

3.2. Dataset

This subsection discusses the three datasets selected for training and validating the model: CICIDS2017, UNSW-NB15, and BoT-IoT. These datasets were chosen for their capacity to represent a variety of relevant attacks and real-world conditions in IoT environments. Each dataset possesses unique characteristics that allow us to evaluate different aspects of network security, providing a solid foundation for the supervised learning of the model.

CICIDS2017, developed by the Canadian Institute for Cybersecurity, is a widely used dataset for network security research, including IoT applications. Generated in a simulated network environment over one week, it captures both normal traffic and a range of attacks such as DDoS, Port Scanning, Brute Force, and SQL Injection. The dataset was recorded using tools like Wireshark and includes over 80 features that encompass flow metrics, packet statistics, and protocol details. While originally designed for enterprise networks, the attack types are relevant to IoT environments, making it suitable for training IDS models for IoT contexts. The dataset's imbalanced nature, with attacks being far less frequent than benign traffic, mirrors real-world network conditions where attacks are rare events. This imbalance is essential for training models capable of effective detection even under typical network traffic scenarios.

The UNSW-NB15 dataset, developed by the University of New South Wales using de IXIA PerfectStorm simulator, captures both normal traffic and a variety of attacks (e.g. brute force, SQL Injection, backdoors, DoS) to simulate real-world scenarios in corporate and IoT networks. It contains over 2.5 million records with 49 features, offering a detailed evaluation of network behaviours. Though not specifically for IoT, its recorded attacks are relevant to this context, making it suitable for training an IoT IDS. Like other datasets, UNSW-NB15 is imbalanced, with normal traffic dominating.

Finally, BoT-IoT, developed by UNSW, is specifically designed to study IoT security threats. Generated using the Cooja simulator under the Contiki operating system, it recreates realistic IoT scenarios, combining legitimate traffic with various attacks such as DDoS, scanning, backdoors, and brute force. The dataset contains millions of records across 36 features, including flow metrics and packet statistics, enabling the identification of anomalous patterns in IoT environments.

The selected datasets – CICIDS2017, UNSW-NB15, and BoT-IoT – allow for training and evaluating the model from complementary perspectives. CICIDS2017 and UNSW-NB15 provide a broad view of attacks, while BoT-IoT focuses specifically on IoT devices. Datasets like NSL-KDD or KDD99 (“KDD-99 Dataset”) were not selected due to their outdated nature, which does not reflect the characteristics of modern traffic or the diversity of current attacks, particularly in IoT environments.

4. Experimentation and evaluation

This section provides a comprehensive overview of the procedure followed to evaluate the intrusion detection model in IoT environments. Data preprocessing is first described,

including the methods used to prepare and balance the datasets. The model implementation and training process are then addressed, highlighting the techniques employed to ensure adequate generalization and enhance intrusion detection capabilities.

4.1. Dataset preprocessing

For the CICIDS2017 dataset, an exhaustive analysis was conducted to assess class distribution, which is essential to understand the dataset's structure and balance before proceeding with model training. Figure 2 shows the class distribution of the dataset under three different approaches: the original dataset, the dataset with the majority class ('BENIGN') undersampled, and the dataset with undersampling of the majority class combined with oversampling of the minority classes using SMOTE + Tomek Links.

In the original distribution, it is clear that the 'BENIGN' class dominates with over two million instances, while the attack classes have significantly fewer samples. This imbalance poses a major challenge, as a model trained with these data could become biased towards the majority class, reducing its effectiveness in detecting attack classes. Therefore, balancing techniques were identified as necessary to address this disparity and ensure the model's ability to accurately identify patterns across all classes. Nevertheless, the original dataset was retained to evaluate the IDS model's performance in a realistic scenario, given that malicious traffic volume is significantly lower than normal traffic.

Then, two modifications were applied to the dataset to observe how these adjustments would impact model performance. The first modification involved undersampling the majority class to reduce the number of examples in the 'BENIGN' class. This was done to decrease the disparity between the majority and minority classes (the various attack types), thereby creating a more balanced dataset. By reducing the dominance of the majority class, the model is given more opportunities to learn the distinctive features of the less frequent classes, thereby improving its ability to detect malicious behaviour patterns.

The second modification was more sophisticated. It involved applying undersampling to the majority class along with oversampling the minority classes using SMOTE and

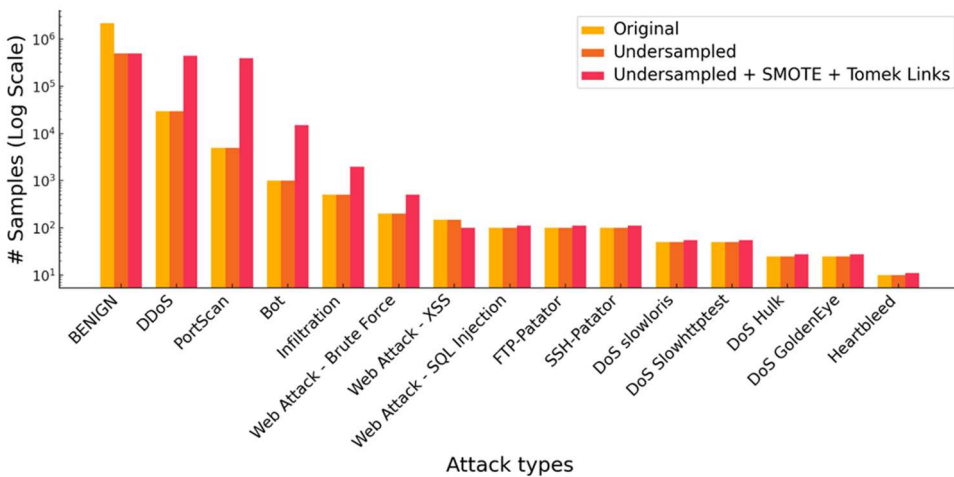


Figure 2. Class distribution in the CICIDS2017 dataset with different preprocessing approaches.

removing ambiguous examples with Tomek Links. SMOTE generates synthetic examples for the minority classes by creating new interpolated data points from existing samples, increasing their representation in the dataset. Tomek Links then eliminate nearby examples from different classes that could introduce noise, thus ensuring that the classes are better defined and separated. This combination aims to enhance the dataset's quality, providing a reasonable balance without fully equating minority and majority classes, as a completely balanced dataset would be unrealistic in network traffic. This approach improves the model's ability to detect attacks while maintaining a practical representation of real-world scenarios.

For the UNSW-NB15 dataset, Figure 3 presents the results of applying the three previously mentioned modifications, now integrated into a single image. Unlike CICIDS2017, UNSW-NB15 classifies the traffic into only two categories: normal traffic (label = 0) and attack traffic (label = 1). This simplification facilitates the detection of malicious activity in a general sense, without differentiating between various types of attacks.

In the integrated figure, the same preprocessing steps applied to the CICIDS2017 dataset are presented, including the original dataset, majority class undersampling, and the combination of undersampling with SMOTE and Tomek Links for the minority classes. The main difference with CICIDS2017 is that in UNSW-NB15, traffic is categorized into only two classes (normal and attack traffic), which simplifies the detection of malicious activity.

For the BoT-IoT dataset, Figure 4 presents the results of the same preprocessing process described for CICIDS2017 and UNSW-NB15, with some key differences specific to BoT-IoT. In this case, the class distribution is distinct, as attack traffic (such as DoS and DDoS) significantly outweighs benign traffic, reflecting the dataset's design to simulate an IoT environment under attack. This imbalance presents a particular challenge, as

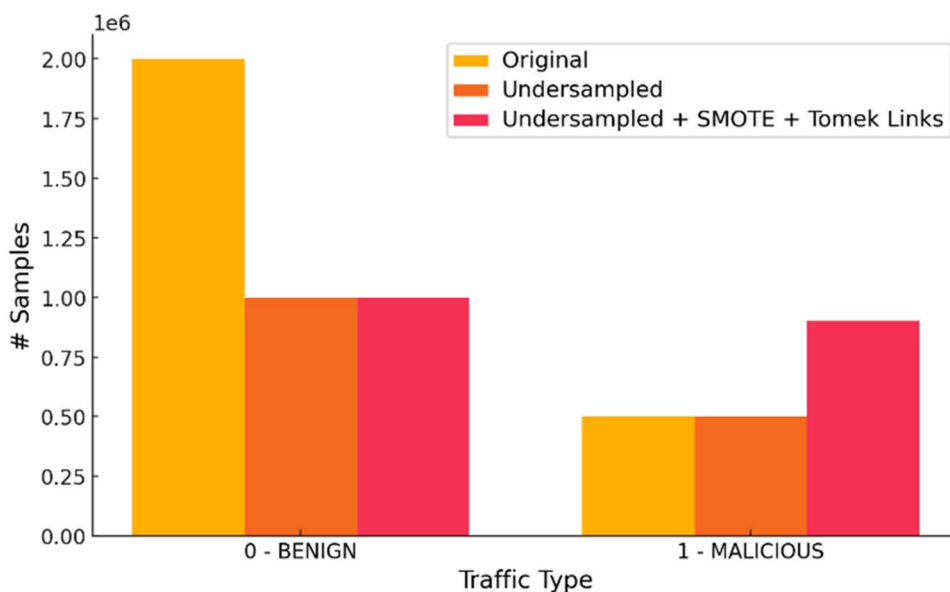


Figure 3. Class distribution in the UNSW-NB15 dataset with different preprocessing approaches.

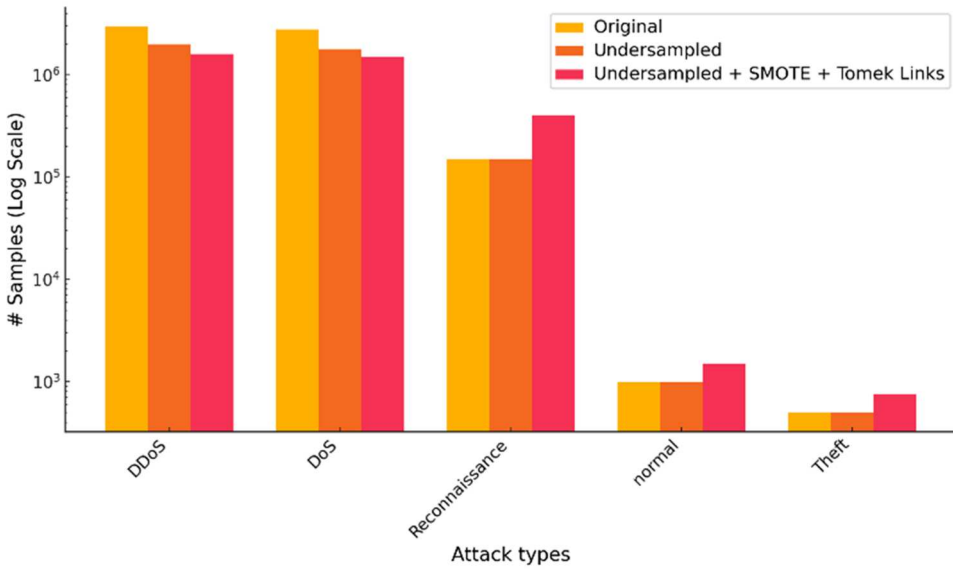


Figure 4. Class distribution in the UNSW-NB15 dataset with different preprocessing approaches.

the overwhelming volume of attack traffic could lead the model to misclassify benign traffic as attacks, increasing the false positive rate.

By applying undersampling to the majority attack classes, the disparity in sample counts was reduced, which improved the model's ability to generalize, preventing overfitting to specific attack features. Finally, combining SMOTE and Tomek Links further increased minority class representation while removing problematic examples, achieving a more balanced dataset. This allowed the model to better distinguish between normal and malicious activities, making it more suitable for real-world IoT scenarios.

4.2. Model training

In this subsection, the training process of the developed DNN model is explained, describing the configurations used and other techniques applied to ensure efficient and reproducible training. The aim is to provide a detailed description that enables replication of this work by following the same steps and configurations.

Before starting the training process, highly correlated features were removed to reduce multicollinearity and avoid redundancy in the feature set. This was done by calculating the correlation matrix for all features and identifying pairs with an absolute correlation value greater than 0.95. Specifically, we used the upper triangle of the correlation matrix to detect such high correlations and selected features to drop accordingly. The identified features were then removed from the dataset, ensuring the model could learn efficiently without being affected by repetitive information, thus reducing the risk of overfitting.

The data was then split into training and testing sets, using 80% for training and 20% for testing. This split was performed in a stratified manner to maintain the original class proportions. Training and testing sets contained representations of each class, even though they remained imbalanced.

The model training was conducted under two different configurations to evaluate performance in various contexts:

- **Without K-Fold:** In this scenario, a conventional method was used to divide the training set into a training subset and a validation subset. This split allows for an evaluation of the model's generalization capacity to unseen data without the need for complex validation methods.
- **With K-Folds:** The model was also trained using the K-Fold cross-validation technique with $K = 10$. This technique divides the training set into ten folds, using nine folds for training and one for validation, rotating the validation fold in each iteration. This way, the model is trained and validated multiple times, providing a more robust evaluation and avoiding dependence on a single data partition. This approach reduces evaluation variance and offers a more representative view of the model's performance, especially in cases of imbalanced datasets.

The use of K-Folds was particularly relevant for evaluating model performance with different data splits, as our goal was to ensure that the results did not depend on a single specific dataset split. However, K-Fold training requires higher computational cost and training time due to the need to train the model multiple times.

The model was trained in the following hyperparameter configurations:

- **Epochs:** A maximum of 50 epochs was used, with Early Stopping halting training if validation loss did not improve for five consecutive epochs to prevent overfitting.
- **Batch size:** A batch size of 1024 was chosen to leverage GPU parallelization, speed up training and optimize memory usage, while balancing against the risk of getting stuck in local minima.
- **Class weights:** Class weights were assigned to give more importance to minority classes and mitigate the impact of dataset imbalances.
- **Callbacks:** Early Stopping and ReduceLROnPlateau were used to control overfitting and improve convergence by adjusting the learning rate when the validation metric plateaued.
- **Optimizer and loss function:** The Adam optimizer with an initial learning rate of 0.0001 was selected for its efficiency in adapting learning rates. Sparse categorical cross-entropy was used as the loss function for multi-class classification.
- **Dropout:** A 30% dropout rate was applied to each dense layer to prevent overfitting by reducing reliance on specific features and ensuring generalization.

During the training and evaluation of the model, metrics such as accuracy, loss, precision, recall, and F1-score were used to assess the model's performance. These metrics provided an understanding of how well the model detected malicious activities and distinguished between different classes. The F1-score was particularly useful for evaluating the model's ability in scenarios with imbalanced datasets. High values of these metrics indicated that the model successfully minimized FP and false negatives (FN), crucial for effective intrusion detection in IoT environments.

5. Results

This section presents the results obtained from training and evaluating the proposed DNN-based IDS model across the different scenarios considered. For each dataset used (CICIDS2017, UNSW-NB15, and BoT-IoT), six different configurations were generated and evaluated. These configurations are divided into two main training approaches: without K-Fold cross-validation and with K-Fold cross-validation ($K=10$). Within each approach, three preprocessing scenarios were tested: the original dataset, the dataset with majority class undersampling to reduce imbalance, and finally, the dataset with both majority class undersampling and SMOTE with Tomek Links for the minority classes. Thus, a total of six results were obtained for each dataset. This exhaustive approach allows for an analysis and comparison of model performance across different training and data balancing contexts.

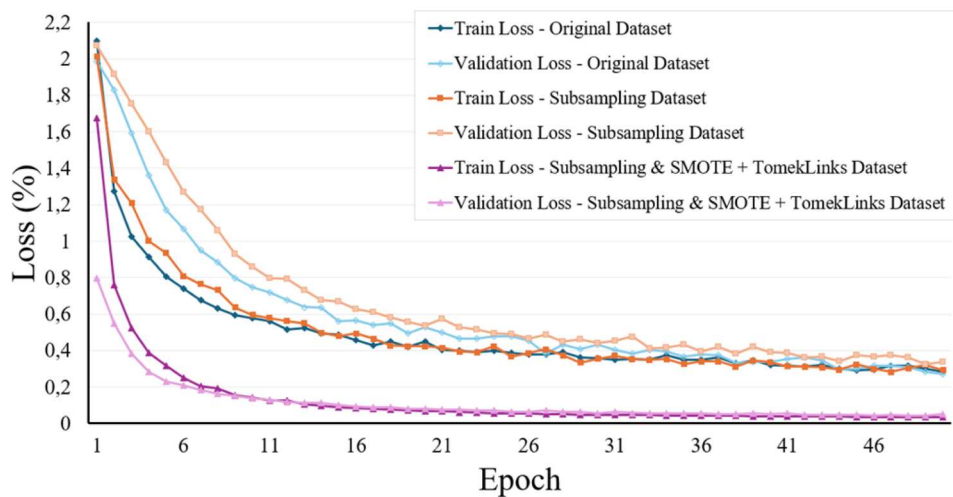
5.1. CICIDS2017

Figure 5 presents the loss and accuracy results during training and validation of the model on the CICIDS2017 dataset, considering the scenario without K-Fold cross-validation. The figure includes the results for three configurations: the original dataset without any balancing, the dataset with majority class undersampling, and the dataset with undersampling combined with SMOTE and Tomek Links. By consolidating these configurations into a single figure, we can better visualize and compare how each preprocessing technique affects the model's learning process.

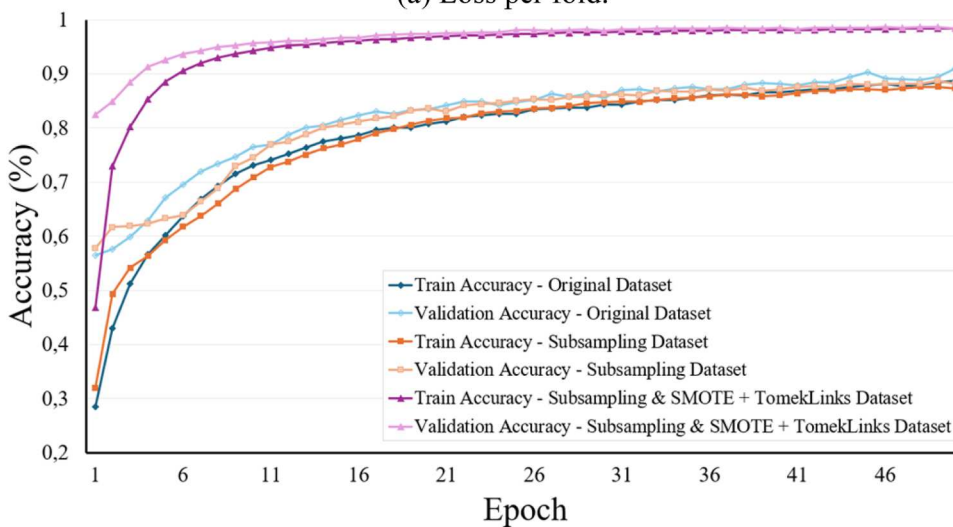
The loss curves for the original dataset show a significant difference between training and validation loss, suggesting challenges in the model's ability to generalize. This discrepancy is likely due to the class imbalance present in the dataset, which favours the majority (BENING) class and prevents effective learning of the minority classes (attack types). This effect is further highlighted by the divergence of accuracy curves, indicating an overfitting tendency where the model performs well on training data but poorly on validation data.

With the use of undersampling (Figure 5), the class imbalance is reduced, which is reflected in the closer alignment of training and validation loss curves, indicating improved generalization. The accuracy curves also show greater stability between training and validation sets, implying a reduced overfitting effect and a better ability of the model to learn from minority class samples. The final configuration, which combines undersampling with SMOTE and Tomek Links, shows the best results. Both loss and accuracy curves exhibit improved convergence, demonstrating that the model benefits from having more balanced classes. The use of SMOTE to generate synthetic samples, combined with Tomek Links to eliminate noisy data, enhances the representation of minority classes, resulting in a model that is more effective and generalizable.

Figure 6 illustrates the results for the same preprocessing configurations, but now with K-Fold cross-validation applied ($k=10$). When comparing Figure 6 with Figure 5, it is evident that K-Folds provides a more robust evaluation process. The loss curves across all configurations show greater consistency and stability, suggesting that training and validating the model multiple times with different data partitions helps mitigate the risk of overfitting and improves the reliability of the evaluation.



(a) Loss per fold.

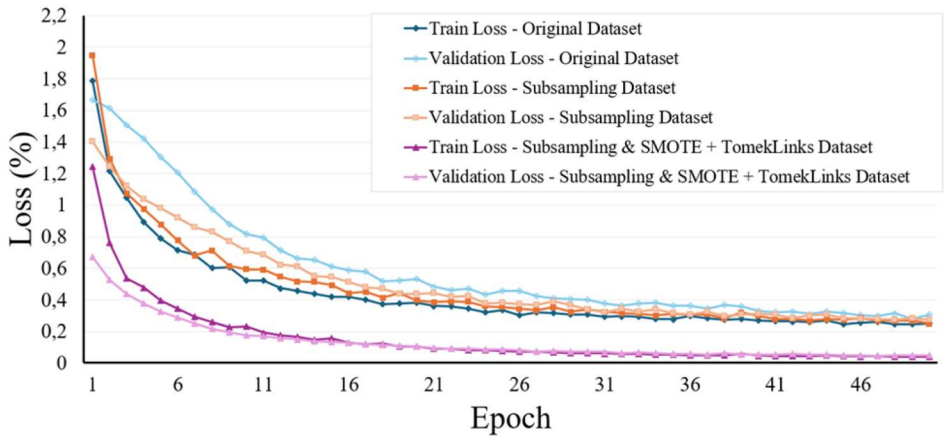


(b) Accuracy per fold.

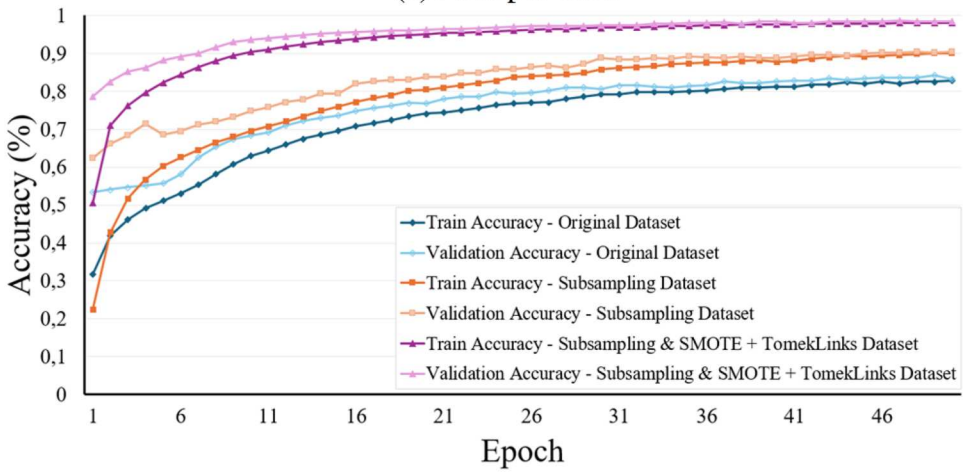
Figure 5. Training results for CICIDS2017 without k-folds. Loss and accuracy curves for both training and validation sets are presented over 50 epochs.

For the dataset with undersampling and SMOTE + Tomek Links, the combination of K-Folds and balanced class representation results in the most consistent and convergent curves. The model demonstrates high accuracy and minimal loss, with a noticeable reduction in overfitting compared to the other scenarios. This reinforces the importance of both data balancing and robust evaluation techniques in achieving reliable intrusion detection in IoT environments.

In the original dataset (Figure 5 (a)), the loss curve shows a considerable difference between training and validation loss, suggesting some difficulty for the model to generalize, likely due to the significant class imbalance. On the other hand, in the dataset with



(a) Loss per fold.



(b) Accuracy per fold.

Figure 6. Training results for CICIDS2017 with k-folds ($k = 10$). Loss and accuracy curves for both training and validation sets are presented over 50 epochs.

undersampling (Figure 5 (b)), the difference between training and validation loss is reduced considerably, indicating an improvement in the model's generalization. Additionally, accuracy in this case exhibits more stability between the training and validation sets, implying that the model is less prone to overfitting and has a better ability to recognize patterns in the minority classes.

Finally, when using undersampling combined with SMOTE and Tomek Links (Figure 5 (c)), greater stability is achieved in both loss and accuracy during training and validation. The loss curves are closer together and show improved convergence, while the accuracy curves indicate that the model has achieved a suitable balance by learning from both majority and minority classes. This confirms that using oversampling techniques like SMOTE, combined with Tomek Links to remove noise, improves minority class representation and helps the model learn in a more effective and generalizable way.

Figures 7 and 8 illustrate how the DNN model performs across the six evaluated scenarios. In the non-K-Folds scenarios (Figure 7), the model demonstrates limited generalization capacity, particularly in the original dataset (Figure 7(a)), where there is a high rate of FN in minority classes, such as ‘Heartbleed’ (label 13) and ‘SQL Injection’ (label 12). This issue is largely due to dataset imbalance, which causes the model to incorrectly predict these underrepresented classes. With the application of undersampling and undersampling combined with SMOTE (Figures 7(b,c)), improvements are seen in the model’s ability to correctly classify minority classes. However, while undersampling and balancing enhance performance, the model still shows some inconsistencies, especially in classifying low-representation classes.

In contrast, the K-Folds scenarios (Figure 8) show a notable improvement in the model’s ability to classify both majority and minority classes. In the original dataset with K-Folds (Figure 8(a)), the number of FN decreases significantly compared to the same scenario without K-Folds. This improvement is attributed to cross-validation, which provides a more robust evaluation, reduces variance, and enables the model to

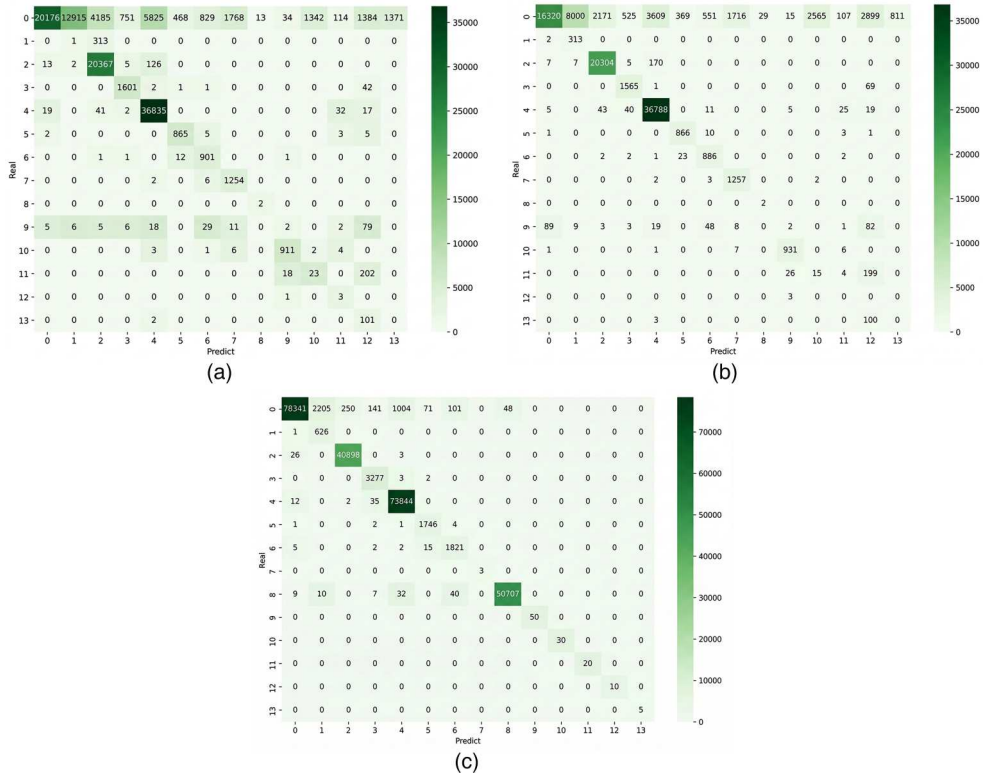


Figure 7. Confusion matrices without cross-validation (k-folds) for the CICIDS2017 Dataset. The matrices display prediction distributions for three configurations: (a) original dataset, (b) undersampling of the majority class, and (c) undersampling with SMOTE + Tomek Links. Diagonal values represent correct predictions, while off-diagonal values indicate errors. Labels: 0 – BENIGN, 1 – DoS Hulk, 2 – PortScan, 3 – DDoS, 4 – DoS GoldenEye, 5 – FTP-Patator, 6 – SSH-Patator, 7 – DoS slowloris, 8 – DoS Slowhttptest, 9 – Bot, 10 – Web Attack Brute Force, 11 – Web Attack XSS, 12 – Web Attack SQL Injection, 13 – Heartbleed.

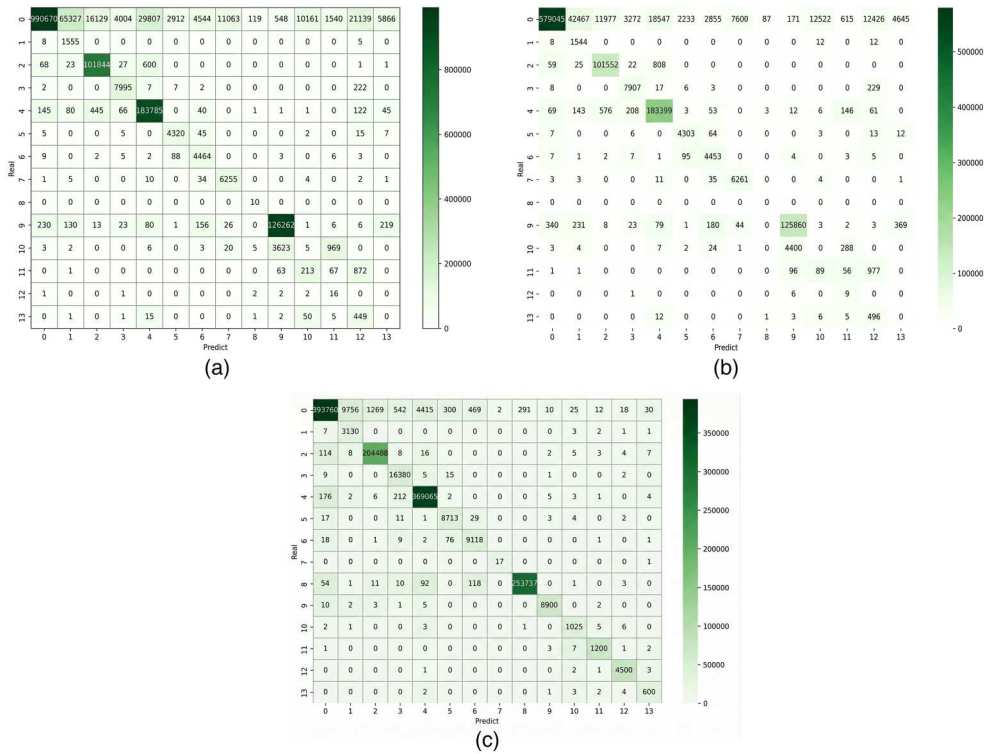


Figure 8. Confusion matrices with cross-validation (k-folds, k=10) for the CICIDS2017 Dataset. The matrices display prediction distributions for three configurations: (a) original dataset, (b) undersampling of the majority class, and (c) undersampling with SMOTE + Tomek Links. Diagonal values represent correct predictions, while off-diagonal values indicate errors. Labels: 0 – BENIGN, 1 – DoS Hulk, 2 – PortScan, 3 – DDoS, 4 – DoS GoldenEye, 5 – FTP-Patator, 6 – SSH-Patator, 7 – DoS slowloris, 8 – DoS Slowhttptest, 9 – Bot, 10 – Web Attack Brute Force, 11 – Web Attack XSS, 12 – Web Attack SQL Injection, 13 – Heartbleed.

learn more general features. The best performance is achieved in the scenario with undersampling and SMOTE combined with K-Folds (Figure 8(c)), where there is a substantial reduction in classification errors, particularly for minority classes. This scenario demonstrates that combining balancing techniques with robust validation like K-Folds enables the model to capture complex patterns in network traffic without overfitting. It also ensures a more balanced representation of classes across training and validation splits.

Table 2 presents the classification report obtained for each evaluated scenario, highlighting key metrics such as precision, recall, and F1-score for each individual class. These metrics provide a detailed assessment of the model's performance in classifying each traffic type, distinguishing between benign traffic and various attack categories.

Analyzing the model's performance across different training scenarios reveals that the best configuration – using K-Folds with subsampling and SMOTE + Tomek Links – achieves an overall accuracy of 99.2%, indicating that the model is highly effective at correctly identifying both normal and malicious traffic. This scenario demonstrates excellent generalization capacity, as reflected in the balance between precision and recall across nearly all evaluated classes.

The high precision in classes such as 'BENIGN' (1.00) in the best-evaluated scenario is primarily due to the large sample size of this class, which helps reduce the number of FP. This consistency is observed across all scenarios, but in those without cross-validation or balancing, precision for classes like 'DoS Hulk' and 'Heartbleed' is considerably lower due to class imbalance, resulting in more FP.

On the other hand, recall for minority classes is also a key metric for understanding how the model handles detections of these specific classes. In unbalanced scenarios, recall for classes like 'DoS Hulk' and 'Heartbleed' is high (1.00), but precision is low, indicating that the model detects most true positives (TP) but also misclassifies many other samples as belonging to these classes, increasing FP. In the best-evaluated scenario, recall remains high while precision improves, suggesting that the model learns more distinctive features that allow it to reduce FP without losing sensitivity.

The F1-score, which balances precision and recall, is particularly relevant for classes like 'DoS Hulk' and 'Heartbleed'. In scenarios with balancing and cross-validation, there is a significant increase in the F1-score for these classes, reaching values above 0.9. This improvement indicates that the model has enhanced its ability to detect accurately (recall) and avoid misclassifications (precision), demonstrating the value of oversampling and cross-validation in improving performance on these minority classes.

In terms of support, classes with higher support, such as 'BENIGN,' generalize well in all scenarios. For classes with lower support, however, the unbalanced scenarios yield lower F1-scores and precision due to the scarcity of examples during training. The use of SMOTE + Tomek Links effectively increases support for these classes, contributing to a notable performance improvement by making the minority classes more representative.

In summary, precision, recall, F1-score, and support metrics in the best scenario illustrate that balancing techniques, combined with cross-validation, enable the model to learn more complex patterns and generalize better, minimizing both FP and FN. This approach is particularly beneficial for improving the detection of lower-support classes, yielding a more stable and accurate performance across the entire CICIDS2017 dataset.

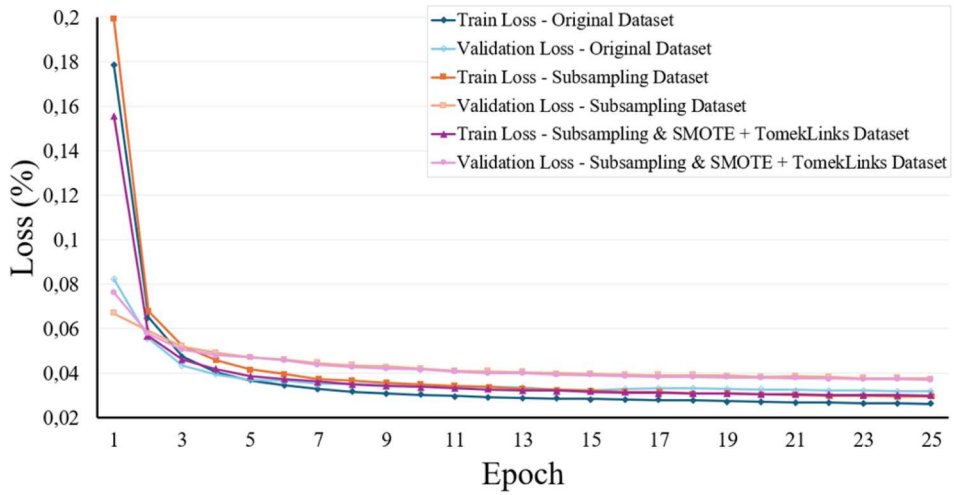
5.2. UNSW-NB15

Figure 9 presents the loss and accuracy results for training and validation of the model on the UNSW-NB15 dataset, without applying K-Fold cross-validation. As with the CICIDS2017 dataset, the figure includes three configurations: the original unbalanced dataset, the dataset with majority class undersampling, and the dataset with undersampling combined with SMOTE and Tomek Links. By consolidating these configurations, it becomes easier to visualize the effect of each preprocessing method on the model's performance.

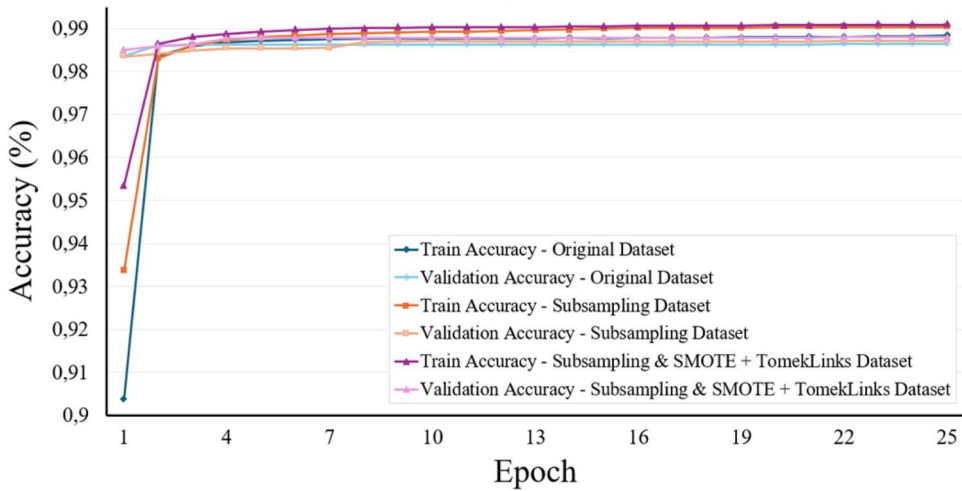
In the original dataset configuration (Figure 9, blue curves), the loss and accuracy results indicate fast convergence, with both training and validation curves stabilizing within the first 25 epochs. This quick stabilization can be attributed to the binary nature of the UNSW-NB15 classification, which is inherently less complex compared to the multi-class setup in CICIDS2017. However, a slight discrepancy remains between the training and validation metrics, indicating potential overfitting due to the class imbalance favouring the majority 'BENIGN' class.

Table 2. CICIDS2017 Classification Report. Presents precision, recall, F1-score, and support for each class across three configurations: the original dataset, the dataset with majority class undersampling, and the dataset with undersampling and balancing through SMOTE + Tomek Links, both without K-Folds and with K-Folds (K = 10). The class labels correspond to: 0 – BENIGN, 1 – DoS Hulk, 2 – PortScan, 3 – DDos, 4 – DoS GoldenEye, 5 – FTP-Patator, 6 – SSH-Patator, 7 – DoS slowloris, 8 – DoS Slowhttptest, 9 – Bot, 10 – Web Attack Brute Force, 11 – Web Attack XSS, 12 – Web Attack SQL Injection, and 13 – Heartbleed.

		Label	0	1	2	3	4	5	6	7	8	9	10	11	12	13
Without K-folds	Original Dataset	Precision	1.00	0.02	0.80	0.58	0.84	0.41	0.27	0.24	0.07	1.00	0.17	0.05	0.00	0.05
		Recall	0.80	0.93	0.99	0.95	0.99	0.97	0.97	0.99	1.00	0.99	0.98	0.05	0.75	0.93
		F1-score	0.89	0.03	0.89	0.72	0.91	0.58	0.42	0.39	0.13	0.99	0.28	0.05	0.00	0.10
		Support	232766	314	20513	1647	36946	880	916	1262	2	25431	927	243	4	105
	Dataset with subsampling over majority class	Precision	1.00	0.03	0.88	0.71	0.90	0.64	0.54	0.46	0.40	1.00	0.25	0.11	0.00	0.08
		Recall	0.82	1.00	0.99	0.96	1.00	0.98	0.97	0.99	1.00	0.99	0.98	0.06	0.80	0.96
		F1-score	0.90	0.06	0.93	0.82	0.95	0.77	0.69	0.63	0.57	0.99	0.40	0.08	0.01	0.15
		Support	139693	315	20493	1635	36936	881	916	1264	2	25429	946	244	4	104
	Dataset with subsampling over majority class and SMOTE over minority classes	Precision	1.00	0.25	1.00	0.96	0.99	0.96	0.96	1.00	1.00	0.92	0.89	0.91	0.87	0.85
		Recall	0.96	1.00	1.00	1.00	1.00	1.00	0.99	1.00	1.00	0.95	0.92	0.93	0.88	0.87
		F1-score	0.98	0.40	1.00	0.98	0.99	0.98	0.97	1.00	1.00	0.94	0.90	0.92	0.87	0.86
		Support	82161	627	40927	3282	73893	1754	1845	3	50805	1094	5204	4786	1573	89
With K-folds	Original Dataset	Precision	0.9995	0.0232	0.8599	0.6593	0.8576	0.5895	0.4806	0.3602	0.0775	0.9956	0.2614	0.1174	0.0007	0.0602
		Recall	0.8512	0.9917	0.9930	0.9709	0.9949	0.9820	0.9742	0.9910	1.0000	0.9930	0.7815	0.1752	0.8000	0.8569
		F1-score	0.9194	0.0453	0.9217	0.7853	0.9211	0.7368	0.6437	0.5284	0.1439	0.9943	0.3917	0.1406	0.0014	0.1124
		Support	1163829	1568	102564	8235	184730	4399	4582	6312	10	127153	4636	1216	20	524
	Dataset with subsampling over majority class	Precision	0.9991	0.0348	0.8899	0.6908	0.9040	0.6479	0.5825	0.4495	0.0938	0.9986	0.2577	0.1240	0.0007	0.0755
		Recall	0.8290	0.9797	0.9911	0.9671	0.9931	0.9762	0.9727	0.9910	1.0000	0.9899	0.9304	0.0730	0.5625	0.9502
		F1-score	0.9062	0.0671	0.9378	0.8059	0.9464	0.7789	0.7289	0.6185	0.1714	0.9942	0.4037	0.0918	0.0014	0.1400
		Support	698462	1576	102466	8176	184679	4408	4578	6318	9	127143	4729	1220	16	522
	Dataset with subsampling over majority class and SMOTE over minority classes	Precision	0.9990	0.2427	0.9937	0.9539	0.9879	0.8950	0.9135	0.9568	0.9367	0.9465	0.8560	0.7800	0.8000	0.9989
		Recall	0.9585	0.9978	0.9993	0.9982	0.9989	0.9820	0.9750	0.9934	0.9885	1.0000	0.9400	0.8200	0.8900	0.9989
		F1-score	0.9783	0.3904	0.9965	0.9756	0.9934	0.9365	0.9435	0.9748	0.9619	0.9444	0.8965	0.8000	0.8415	0.9989
		Support	410804	3137	204634	16409	369463	5222	4891	8771	9242	17	3324	1243	37	254023



(a) Loss per fold.



(b) Accuracy per fold.

Figure 9. Training results for UNSW-NB15 without k-folds. Loss and accuracy curves for both training and validation sets are presented over 50 epochs.

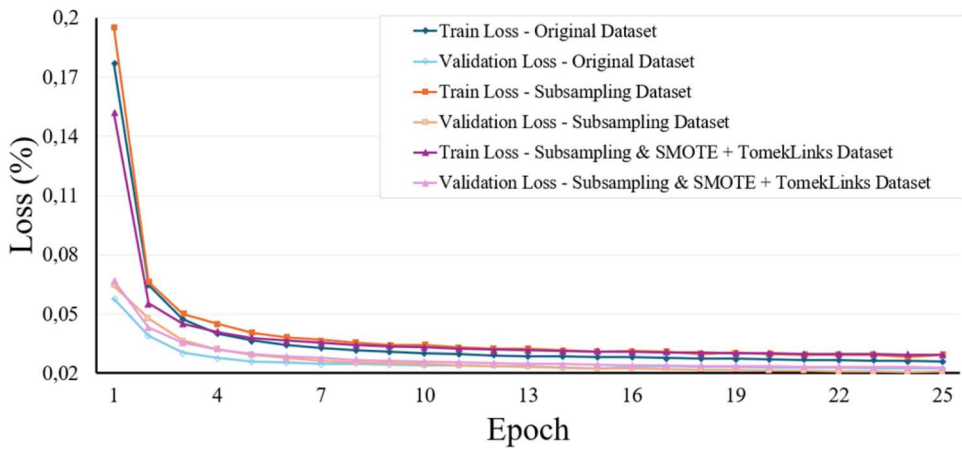
When using undersampling (orange curves in Figure 9), the results show an improvement in generalization, evidenced by the closer alignment of loss and accuracy curves between training and validation. This similarity implies a reduced overfitting tendency, as the model now learns more evenly from both classes, minimizing the bias towards the majority class. This effect is similar to the improvement observed in the CICIDS2017 dataset, although the binary classification task in UNSW-NB15 allows for a more stable overall performance.

The configuration that combines undersampling with SMOTE and Tomek Links (pink curves in Figure 9) exhibits the most balanced outcome. The improved convergence of both loss and accuracy curves suggests that the combination of synthetic data generation

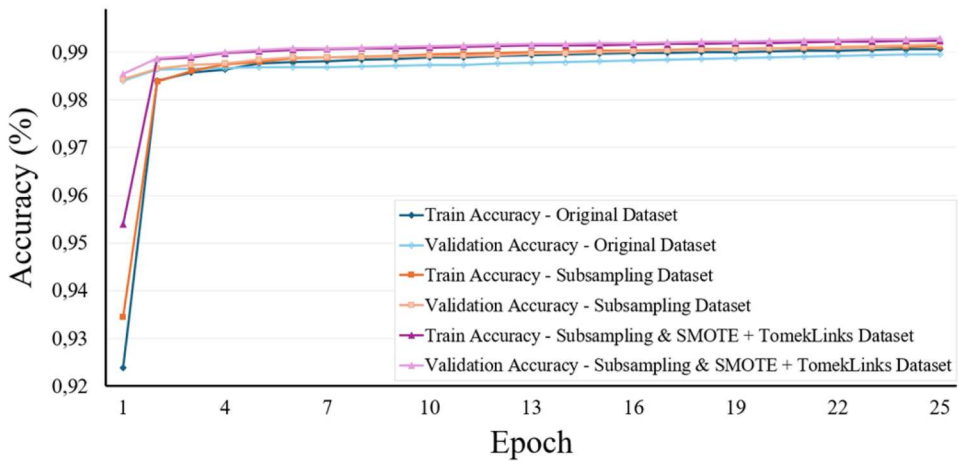
and noise reduction successfully addresses the imbalance, allowing the model to generalize effectively without significant overfitting. Similar to the results in CICIDS2017, the use of SMOTE to enhance minority class representation proves to be crucial for reliable detection of malicious traffic.

Figure 10 presents the same configurations but includes K-Fold cross-validation ($k = 10$). In comparison to Figure 9, the application of K-Folds significantly improves the stability and consistency of the model's performance across all scenarios. This is evident from the smoother curves for both loss and accuracy, indicating that training on multiple data partitions allows the model to better capture the underlying data distribution, thereby enhancing generalization.

For the original dataset configuration (blue curves in Figure 10), the K-Fold approach helps reduce the slight gap between training and validation metrics seen in Figure 9. The



(a) Loss per fold.



(b) Accuracy per fold.

Figure 10. Training results for UNSW-NB15 with k-folds ($k = 10$). Loss and accuracy curves for both training and validation sets are presented over 50 epochs.

binary nature of the classification makes the learning task simpler, but cross-validation ensures that the model is exposed to varied training data, which mitigates overfitting risks and provides a more robust evaluation.

In the undersampled configuration (orange curves in Figure 10), the K-Fold cross-validation further enhances stability, minimizing fluctuations in loss and accuracy between training and validation sets. This result mirrors the CICIDS2017 findings, where K-Fold cross-validation helped balance the learning process across multiple data segments, ultimately reducing the variance caused by undersampling.

Finally, for the combined undersampling and SMOTE + Tomek Links configuration (pink curves in Figure 10), the results are consistent with what was observed in CICIDS2017. The cross-validation process, when combined with data balancing, produces the most stable and overlapping training and validation curves, demonstrating effective generalization. The binary nature of UNSW-NB15 simplifies the classification task, making the improvements in accuracy and loss convergence more pronounced compared to CICIDS2017, where the multi-class problem introduced greater complexity. Comparing UNSW-NB15 (Figures 9 and 10) with CICIDS2017 (Figures 5 and 6), we see that UNSW-NB15 consistently achieves faster and more stable convergence. This is largely due to the binary classification setup, which simplifies the learning process and reduces the computational burden compared to the multi-class classification in CICIDS2017.

While both datasets benefit from the application of data balancing techniques and cross-validation, the effect is more significant in CICIDS2017 due to the greater challenge posed by the multiple attack classes. For UNSW-NB15, the convergence and generalization are achieved with fewer epochs, as evidenced by the early stopping after 25 epochs in both training scenarios.

This contrasts with CICIDS2017, which required all 50 epochs to converge, particularly in configurations involving class balancing.

In both datasets, the combination of SMOTE, Tomek Links, and K-Fold cross-validation proves to be the most effective strategy for achieving balanced learning and reducing overfitting. However, the impact is more pronounced in CICIDS2017 due to the complexity of the dataset. For UNSW-NB15, which involves simpler binary classification, these methods contribute to a highly stable model with minimal overfitting, making it well-suited for real-world IoT environments where efficient and reliable threat detection is critical.

Figures 11 and 12 present the confusion matrices for the UNSW-NB15 dataset under different preprocessing scenarios, with and without K-Fold cross-validation. Figure 11 shows results without K-Fold, illustrating the challenges in detecting malicious traffic, particularly in the original dataset configuration, with high numbers of FN. However, balancing methods like undersampling and SMOTE + Tomek Links significantly improve classification, particularly reducing FN for malicious traffic. In contrast, Figure 12 demonstrates the effect of K-Fold cross-validation, where the model's ability to accurately classify malicious traffic improves across all configurations. Notably, the combination of K-Fold with SMOTE and Tomek Links achieves the best results, reflecting the robustness of the evaluation process and enhanced learning of minority classes.

Without K-Fold cross-validation (Figure 11), the original dataset (Figure 11(a)) shows significant classification challenges, especially for malicious traffic, resulting in many FN. However, class balancing techniques improve the model's performance in detecting

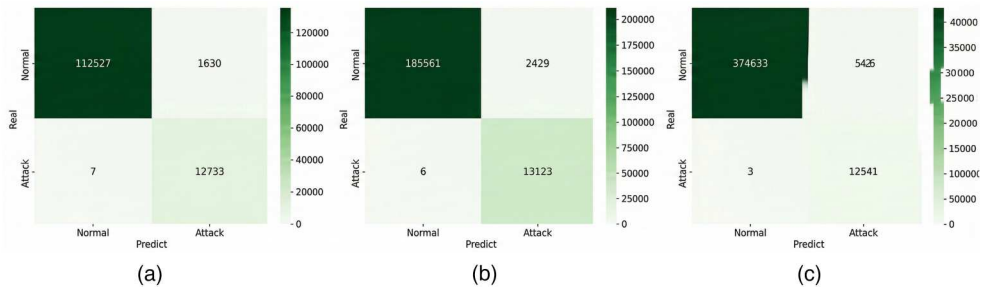


Figure 11. Confusion matrices without cross-validation for the UNSW-NB15 Dataset. Confusion matrices showing classification results for three configurations: (a) original dataset, (b) majority class undersampling, and (c) undersampling with SMOTE + Tomek Links. Diagonal cells indicate correct predictions, while off-diagonal cells represent errors. Labels: 0 – Normal traffic, 1 – Malicious traffic.. (a) Original Dataset. (b) Dataset with subsampling on majority class. (c) Dataset with subsampling on majority class and SMOTE + Tomek Links over minority classes.

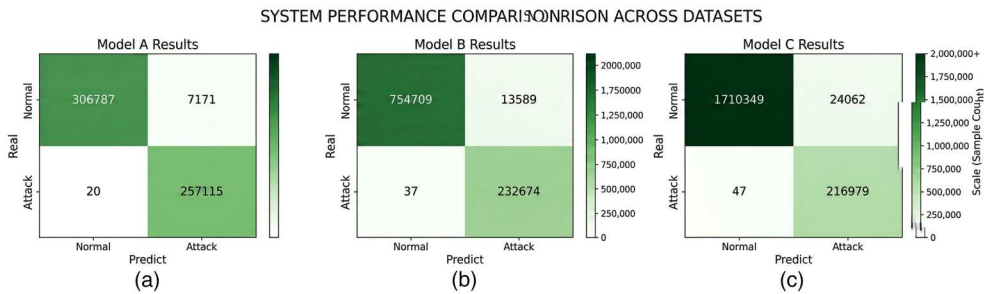


Figure 12. Confusion matrices with cross-validation (k-Folds, k = 10) for the UNSW-NB15 Dataset. Confusion matrices for three configurations: (a) original dataset, (b) majority class undersampling, and (c) undersampling with SMOTE + Tomek Links, using K-Folds. Diagonal cells show correct predictions, while off-diagonal cells represent errors. Labels: 0 – Normal traffic, 1 – Malicious traffic. These matrices assess the model's ability to differentiate between normal and attack traffic in each configuration.

malicious traffic, reducing FN in balanced scenarios (Figures 11(b,c)). Applying K-Fold cross-validation (Figure 12) further enhances the model's ability to differentiate between normal and malicious traffic, particularly reducing FN in both unbalanced (Figure 12(a)) and balanced configurations (Figures 12(b,c)).

Comparing UNSW-NB15 (Figures 11 and 12) with CICIDS2017 (Figures 7 and 8), UNSW-NB15 consistently achieves better results, particularly with fewer FN. This improvement is attributed to the simpler binary classification of UNSW-NB15, which allows the model to learn more efficiently compared to the multi-class setup of CICIDS2017. The reduced complexity leads to fewer errors and better overall performance in distinguishing attack types from normal traffic.

The best results are observed in the UNSW-NB15 dataset with SMOTE balancing and K-Fold cross-validation (Figure 12(c)), which achieves the highest accuracy and optimal detection of malicious traffic with minimal errors. This scenario highlights the advantage of combining K-Fold and balancing techniques, especially in simpler classification tasks

like UNSW-NB15, which surpasses the performance achieved on CICIDS2017 even with the same methods.

Table 3 presents the classification report for the UNSW-NB15 dataset in all evaluated scenarios, summarizing key metrics such as precision, recall, and F1-score for each configuration. Unlike CICIDS2017, which included multiple attack categories, UNSW-NB15 poses a binary classification challenge. This report allows for an assessment of the model's ability to effectively distinguish between these two categories. Additionally, the support metric is included, representing the number of instances per class and providing valuable insight into the model's performance in differentiating between normal and attack traffic, especially under various data balancing techniques and cross-validation strategies.

Observing the model's performance across different UNSW-NB15 scenarios confirms that the optimal configuration is the use of K-Fold combined with undersampling and SMOTE + Tomek Links. This setup achieves high levels of precision and recall, maximizing the model's generalization capacity and minimizing classification errors. For UNSW-NB15, results show near-100% accuracy for both normal and attack traffic, highlighting the model's effectiveness in binary classification.

As noted in the analysis of CICIDS2017, balancing techniques and cross-validation are essential for enhancing the model's generalization capability, and this is similarly reflected in UNSW-NB15. In scenarios without K-Fold, the gap between precision and recall is more pronounced, especially for attack traffic, where a drop in precision suggests a higher

Table 3. UNSW-NB15 classification report. Presents precision, recall, F1-score, and support for each class across three configurations: the original dataset, the dataset with majority class undersampling, and the dataset with undersampling and balancing through SMOTE + Tomek Links, both without K-Folds and with K-Folds (K = 10). The class labels correspond to: 0 – Normal, 1 – Attack.

		Label	0	1
Without K-folds	Original Dataset	Precision	1.00	0.8860
		Recall	0.9856	0.9997
		F1-score	0.9928	0.9394
		Support	114167	12740
	Dataset with subsampling over majority class	Precision	1.00	0.9300
		Recall	0.9855	0.9998
		F1-score	0.9927	0.9636
		Support	167990	32259
	Dataset with subsampling over majority class and SMOTE over minority classes	Precision	1.00	0.6992
		Recall	0.9857	0.9998
		F1-score	0.9928	0.8229
		Support	379471	12594
With K-folds	Original Dataset	Precision	1.00	0.9728
		Recall	0.9860	0.9999
		F1-score	0.9929	0.9862
		Support	513944	257135
	Dataset with subsampling over majority class	Precision	1.00	0.9731
		Recall	0.9862	0.9999
		F1-score	0.9930	0.9863
		Support	765278	382711
	Dataset with subsampling over majority class and SMOTE over minority classes	Precision	1.00	0.9124
		Recall	0.9861	0.9998
		F1-score	0.9930	0.9541
		Support	1775011	257026

incidence of FP. However, with the inclusion of K-Fold, a better balance between precision and recall is achieved, reducing variability in classification outcomes.

As in CICIDS2017, the accuracy for the benign traffic class remains near 1.00 across all scenarios, which is attributable to the large number of samples in this class. This high support enables the model to learn representative patterns of benign traffic, thus reducing the FP rate. In contrast, the attack traffic class has lower support, which can decrease the model's ability to generalize accurately in unbalanced scenarios due to the scarcity of examples. However, applying SMOTE increases the support for the attack class, enhancing its representativeness and contributing to a significant improvement in precision, recall, and F1-score.

Cross-validation also enhances the model's stability in classifying the attack class, significantly reducing FP and FN. In particular, the configuration with K-Fold and SMOTE balancing achieves a higher and more stable F1-score for the attack class, indicating that the model not only has a strong capacity to detect attacks (high recall) but also avoids misclassifications (improving precision), achieving a better balance between these metrics.

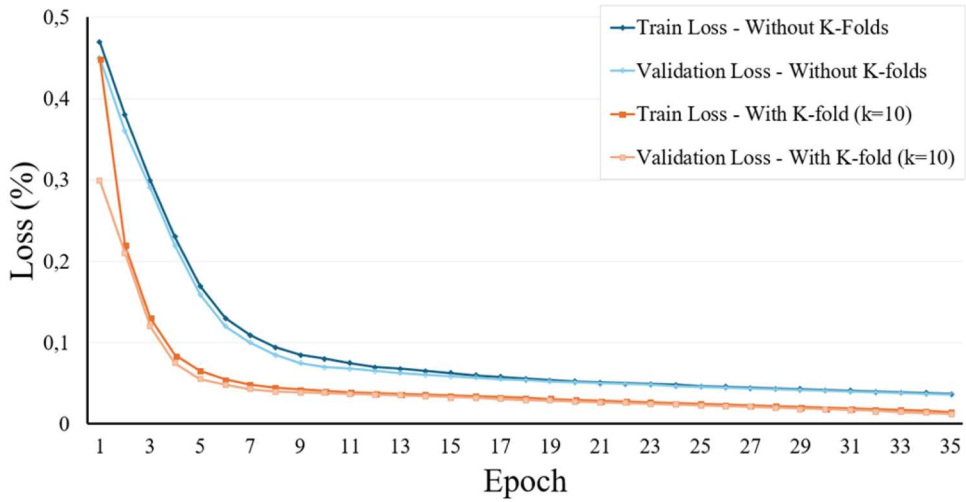
5.3. BoT-IoT

In this subsection, we focus on analyzing the model's performance on the BoT-IoT dataset, applying the configuration that achieved the best results with the CICIDS2017 and UNSW-NB15 datasets: majority class undersampling combined with SMOTE + Tomek Links for the minority class, along with K-Fold cross-validation. This choice is grounded in prior findings, where this configuration significantly enhanced the model's generalization capability, yielding high precision, recall, and F1-scores while minimizing classification errors and maximizing stability.

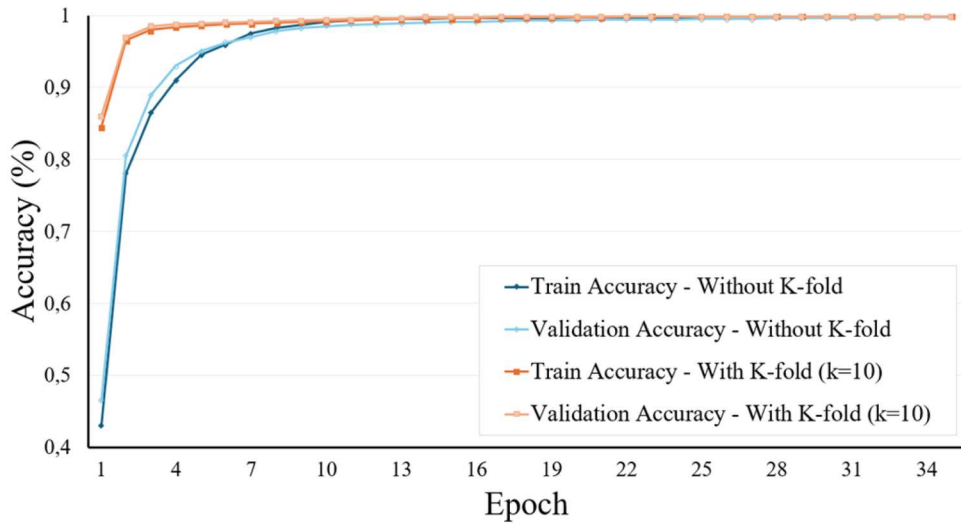
The BoT-IoT dataset presents unique characteristics compared to CICIDS2017 and UNSW-NB15. Specifically designed to represent network traffic in IoT environments, which are typically resource-constrained and prone to distinct vulnerabilities, BoT-IoT introduces challenges that are different from the other datasets. The primary goal here is to determine whether the balancing and validation techniques that were successful for other datasets can also yield effective results in the context of IoT-specific network traffic.

Figure 13 presents the loss and accuracy curves for scenarios both with and without cross-validation (K-Folds, $k = 10$) applied to the BoT-IoT dataset. In the scenario without cross-validation (Figure 13(a)), we observe a good convergence of the loss and accuracy curves for both training and validation phases. However, there is a small gap between these curves, indicating a slight overfitting tendency where the model performs well during training but shows reduced consistency during validation. When K-Fold is applied (Figure 13(b)), the curves demonstrate greater consistency, with minimal differences between training and validation, reflecting a substantial improvement in the model's generalization capability.

Figure 14 presents confusion matrices for scenarios with and without cross-validation, using the configuration with undersampling and SMOTE + Tomek Links. In the scenario without K-Fold (Figure 14(a)), several misclassifications are observed, particularly for the 'Reconnaissance' and 'Theft' classes, leading to an increased number of false positives (FP) and false negatives (FN). This suggests that the model struggles to effectively



(a) Loss per fold.



(a) Accuracy per fold.

Figure 13. Training results for BoT-IoT. Loss and accuracy curves for both training and validation sets are presented over 50 epochs.

distinguish these classes, especially when there are fewer representative examples available. In contrast, when K-Fold is applied (Figure 14(b)), there is a notable improvement, with an increase in true positives (TP) and a significant reduction in both FP and FN across all classes. This enhancement illustrates the impact of cross-validation in enabling the model to robustly learn distinctive patterns of different traffic types, thereby improving its generalization capabilities.

Comparing these results with those obtained from the other datasets, BoT-IoT stands out due to its distinct traffic composition of benign and malicious data. Although BoT-IoT contains fewer categories of attack compared to CICIDS2017, it has a significantly larger

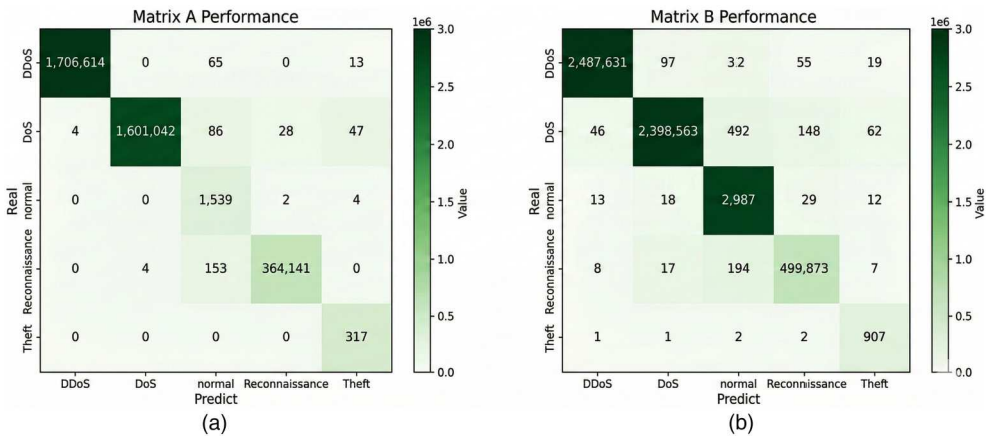


Figure 14. Confusion matrices for SMOTE + Tomek Links and subsampling configurations, comparing results with and without cross-validation. The matrices illustrate the classification performance across five classes: DDoS, DoS, normal, Reconnaissance, and Theft, showing true positives along the diagonal and misclassifications in off-diagonal cells.

volume of malicious traffic and a very limited amount of benign traffic, presenting a considerable challenge for effective classification. CICIDS2017, by contrast, included more attack classes, which added complexity to the classification process but had a more balanced class distribution, which aided in differentiating specific attacks from benign traffic.

In the case of UNSW-NB15, the problem was simplified to a binary classification, enabling faster and more stable model convergence. However, in BoT-IoT, the higher proportion of malicious traffic and the limited benign traffic introduced additional challenges for avoiding misclassifications. Nevertheless, the application of K-Folds and balancing techniques using SMOTE + Tomek Links provided performance that was comparable to other datasets, with clear improvements in reducing FP and FN.

Overall, the combination of K-Fold cross-validation and class balancing techniques proved to be the most effective strategy for the BoT-IoT dataset, providing strong generalization capabilities and minimizing classification errors, particularly in the 'Reconnaissance' and 'Theft' classes. Despite the unique characteristics of BoT-IoT that differentiate it from the other datasets, the appropriate use of cross-validation and balancing techniques enabled the model to adapt well to the dataset's characteristics, achieving satisfactory results and high accuracy, similar to the best outcomes observed with CICIDS2017 and UNSW-NB15.

Finally, BoT-IoT's unique traffic distribution – with a limited number of benign samples and an overwhelming proportion of malicious traffic – introduces significant challenges for traditional classification methods. Standard training approaches tend to risk bias towards over-detecting attack instances. In contrast, the configuration using undersampling, SMOTE + Tomek Links, and K-Fold cross-validation successfully counteracts this imbalance. This approach prevents overfitting on majority attack classes while maintaining sensitivity across all classes, resulting in a well-balanced and effective detection model.

Further optimization of this configuration could be beneficial for other IoT datasets. The characteristics of BoT-IoT suggest that adjusting balancing parameters, such as the

Table 4. BoT-IoT classification report. Presents precision, recall, F1-score, and support for each class of the dataset with undersampling and balancing through SMOTE + Tomek Links, both without K-Folds and with K-Folds (K = 10). The class labels correspond to: 0 – DDoS, 1 – DoS, 2 – Normal, 3 – Reconnaissance, 4 – Theft.

		Label	0	1	2	3	4
Without K-folds	Dataset with subsampling over majority class and SMOTE over minority classes	Precision	0.9865	0.9789	0.8025	0.9854	0.8654
		Recall	0.9865	0.9699	0.9551	0.9555	0.8655
		F1-score	0.9953	0.9261	0.8605	0.9712	0.8565
		Support	152392	25465	953	2541	117
With K-folds	Dataset with subsampling over majority class and SMOTE over minority classes	Precision	1.00	1.00	0.8217	0.9999	0.8320
		Recall	1.00	0.9999	0.9961	0.9995	1.00
		F1-score	1.00	0.9961	0.9005	0.9997	0.9083
		Support	1706692	1601207	1545	364328	317

undersampling ratio or SMOTE ratios, could enhance performance for extremely imbalanced datasets. Future work could explore adaptive balancing methods that dynamically adjust these parameters based on dataset-specific distributions, enhancing the model's adaptability and performance across diverse IoT environments.

Table 4 presents the classification report for BoT-IoT under two configurations: one without K-Fold and another with K-Fold cross-validation, both using undersampling on the majority class and SMOTE + Tomek Links for the minority classes. In the non-K-Fold scenario, a high precision is observed in the majority class 'DDoS' (0.9865) and the 'Normal' class (0.9854). However, for less represented attack classes, such as 'Reconnaissance' and 'Theft' (both around 0.865), precision and recall values are notably lower. This indicates that, although the balancing technique improves the model's performance, minority classes remain a challenge due to their low support. The F1-score also reflects this disparity, being lower for classes with less representation.

With the addition of K-Fold cross-validation, overall performance and consistency across classes improve. Precision and recall for the majority 'DDoS' class reach 100%, indicating the model's strong generalization ability in this configuration. Additionally, the F1-score for all classes shows significant improvements, highlighting an enhanced ability to detect and accurately classify different traffic categories. Specifically, the 'Reconnaissance' class shows an increase in F1-score, evidencing the model's improved effectiveness in learning from minority classes, thanks to cross-validation.

As seen with the CICIDS2017 and UNSW-NB15 datasets, K-Fold cross-validation combined with balancing techniques like SMOTE and undersampling proves to be the optimal configuration for achieving high levels of precision, recall, and F1-score. This setup allows the model to better generalize and minimize both FP and FN, especially in classes with lower support, where cross-validation is essential for stabilizing and enhancing model performance.

Table 5 presents a summary of related studies, highlighting the techniques used, datasets employed, and results obtained. Our proposed model compares favourably with other approaches, showing competitive results, particularly in terms of precision and generalization across different datasets.

Many previous studies have achieved good results in terms of precision and F1-score using architectures like LSTM, GRU, and combinations such as CNN-LSTM with GAN. However, our approach focuses on enhancing the model's ability to handle diverse IoT

Table 5. Summary of IDS model comparisons for IoT environments. Presents a summary of different IDS models evaluated in IoT environments, including the datasets used, preprocessing techniques applied, and the obtained results.

Article	IDS Model	Dataset Used	Preprocessing	Results Obtained
Almiani et al., 2020	Deep RNN for Fog Computing	NSL-KDD	Symbolic to numerical, feature reduction, SMOTE	Precision 92.42%, FP 10.06%, MCC 0.8496. Improved detection for complex attacks (U2R; R2L)
Ullah et al., 2023	Transformer-based IDS	MQTT-IoT-IDS2020	Feature selection, normalization	Accuracy 99.99%, F1-Score 99.99%
Soliman et al., 2023	LSTM, Bi-LSTM, GRU	ToN_IoT	Data cleaning, normalization, SMOTE, SVD	Binary: BiLSTM 99.999%, LSTM 99.991%, GRU 99.999%. Multiclass: BiLSTM 99.9%, LSTM 99.908%, GRU 99.984%
Li et al., 2024	CNN-LSTM with GAN	NSL-KDD, CICIDS2018, BoT-IoT	One-Hot Encoding, SMOTE, Z-Score normalization	NSL-KDD: Precision 99.63%, Recall 99.71%, F1 99.66%. CICIDS2018: Precision 98.75%, Recall 98.85%, F1 98.82%
Swathi & Bindu, 2024	LSTM Encoder-Decoder	Time series datasets	Z-Score normalization, PCA, segmentation	Precision 92-100%, Recall 0.04-0.08
Osman et al., 2024	Ensemble Learning + GA	RPL-ELIDS	Data cleaning, normalization, GA for feature selection	Precision: 99.18% (VN), 99.38% (DR), 99.66% (DIS), overall multiclass 97.9%
Ramana et al., 2022	GRU + WOA	WSN-DS	Normalization, multiclass labelling	Accuracy 99.85%
Nie et al., 2024	Multi-task Learning	BoT-IoT, IoT Network Intrusion	Multi-view representation, normalization	BoT-IoT: Precision 99.76%, Recall 99.84%. IoT Network Intrusion: Precision 98.31%, Recall 98.93%
Guo et al., 2020	GGM-VAE Autoencoder	Sensor data (Intel, Yahoo)	Interpolation, Z-Score normalization	Yahoo: Precision 98.5%, F1-Score 95.7%
Ahmad et al., 2021	Decision Trees (RF, SVM, ANN)	UNSW-NB15	Imputation, class imbalance handling	Binary: RF 98.67%, SVM 97.69%, ANN 94.78%. Multiclass: RF 97.37%
Our model	DNN with/without K-Fold CV	CICIDS2017, UNSW-NB15, BoT-IoT	Data cleaning, oversampling, normalization	CICIDS2017: Precision 99.2%, Recall 98.9%, F1 99.0%. UNSW-NB15: Precision 98.7%, Recall 97.4%, F1 98.0%. BoT-IoT: Precision 99.4%, Recall 99.1%, F1 99.2%

traffic, particularly in detecting infrequent attacks, by effectively integrating advanced data validation and balancing techniques.

The proposed model distinguishes itself by emphasizing stability and consistent performance across representative datasets (CICIDS2017, UNSW-NB15, and BoT-IoT). This focus on improving minority class detection and reducing classification errors reflects a more holistic approach compared to previous works, which often prioritize a single metric or specific configurations.

Rather than merely pursuing the highest accuracy, our approach underscores the importance of robustness and the model's ability to maintain reliable performance across varied traffic conditions. This broader perspective provides a notable advantage in terms of applicability to IoT security, where adaptability and resilience are crucial in the face of diverse and complex network traffic.

6. Conclusion

The study demonstrates that the optimal configuration for maximizing the generalization capacity of an IDS model in IoT environments involves combining subsampling, SMOTE +

Tomek Links, and K-Fold cross-validation. This configuration resulted in accuracies of 99.2% for CICIDS2017, 99.7% for UNSW-NB15, and 99.8% for BoT-IoT, with the differences in accuracy reflecting the varying complexities of the datasets. While CICIDS2017 presented a challenging multiclass classification problem, UNSW-NB15 simplified the task with a binary classification setup, and BoT-IoT, despite being multiclass, benefited from a predominance of malicious traffic that improved model stability.

In addition to achieving high accuracy, this study provides a comprehensive evaluation of the model under three different data preprocessing configurations and the impact of using K-Fold cross-validation versus not using it. The results show that K-Fold cross-validation is essential to reduce overfitting and ensure robust evaluation. Compared to previous approaches, such as CNN for ToN_IoT (Soliman et al., 2023) or GA (Osman et al., 2024) for feature selection, the proposed DNN model not only achieves competitive accuracy but also offers improved adaptability and efficiency for resource-constrained IoT devices. This dual contribution underscores the viability of the proposed solution, balancing model simplicity, adaptability, and performance while ensuring reliable generalization.

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